

Customer Interface Box Logic Unit Alarm Indication Not Received When Alarm Activated

Symptoms

- No communication between the DCU410 unit and CLU unit when alarm is activated.

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

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Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the customer interface box.	
	STEP 1A. Check the DCU410 unit display for faults.	
	STEP 1A-1. Check the CLU power supply wire for voltage +24-VDC.	
	STEP 1B. Check the customer interface box logic unit alarm signal wire for an open.	
	STEP 1C. Check the customer interface box logic unit alarm signal wire for a wire to wire short.	
	STEP 1D. Check the customer interface box logic unit alarm signal wire for short to ground.	

STEP 1. Check the customer interface box.

STEP 1A. Check the DCU410 unit display for faults.

Conditions :

- Locate the DCU410 unit display.

Action	Specification/Repair	Next Step
Check the DCU410 unit display for indication of faults.	DCU410 unit indicates fault(s)? YES	1B
	DCU410 unit indicates fault(s)? NO	1A-1

STEP 1A-1. Check the CLU power supply wire for voltage +24-VDC.

Conditions :

- Open the customer interface box.

Action	Specification/Repair	Next Step
Check the voltage at the battery 1 voltage (switched power) at the DCU410 unit. • Place one test lead on the battery 1 voltage supply wire at the CLU. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than +24-VDC. YES Repair : Check the batteries. Refer to the OEM service manual.	Repair complete
	Less than +24-VDC. NO	1B

STEP 1B. Check the customer interface box logic unit alarm signal wire for an open.

Conditions :

- Open the customer interface box.
- Disconnect the customer interface box logic unit alarm signal wire at the DCU410 unit and CLU.

Action	Specification/Repair	Next Step
Check the customer interface box logic unit alarm signal wire for an open. - Place one test lead on the customer interface box logic unit signal wire at the DCU410 unit. - Place the other test lead on the customer interface box logic unit signal wire at the CLU X4 connection. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	1C
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete

STEP 1C. Check the customer interface box logic unit alarm signal wire for a wire-to-wire short.

Conditions : Open the customer interface box.		
Action	Specification/Repair	Next Step

Check the customer interface box logic unit alarm signal wire for a wire-to-wire short. - Place one test lead on the customer interface box logic unit alarm signal wire at the DCU410 unit. - Place the other test lead on all other wires at the DCU410 unit. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html) Replace the CLU. Contact a Cummins® Authorized Repair Location.	Repair complete
	Less than 10 ohms? NO	1D

STEP 1D. Check the customer interface box logic unit alarm signal wire for a short to ground.

Conditions : Open the customer interface box.		
Action	Specification/Repair	Next Step
Check the customer interface box logic unit alarm signal wire for a short to ground. - Place one test lead on the customer interface box logic unit alarm signal wire at the DCU410 unit. - Place the other test on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html) Replace the CLU. Contact a Cummins® Authorized Repair Location.	Repair complete
	Less than 10 ohms? NO	Contact a Cummins® Authorized Repair Location.

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